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Memo No. _____

Date / /

(56) Equivalence of two Finite Automata

Steps:

(1) For any pair of states $\{q_i, q_j\}$ the transition for input $a \in \Sigma$ is ~~defined~~ defined by $\{q_a, q_b\}$ where $\delta\{q_i, a\} = q_a$ and $\delta\{q_j, a\} = q_b$

The two automata are not equivalent if for a pair $\{q_a, q_b\}$ one is INTERMEDIATE State and the other is FINAL State,

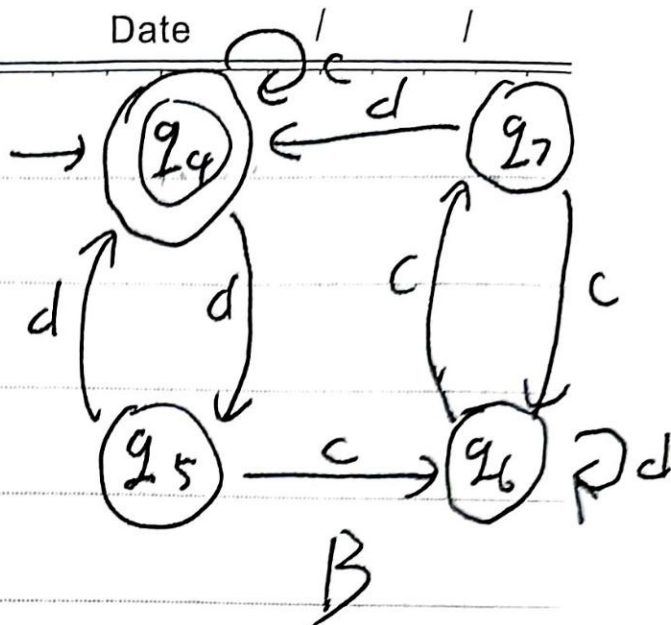
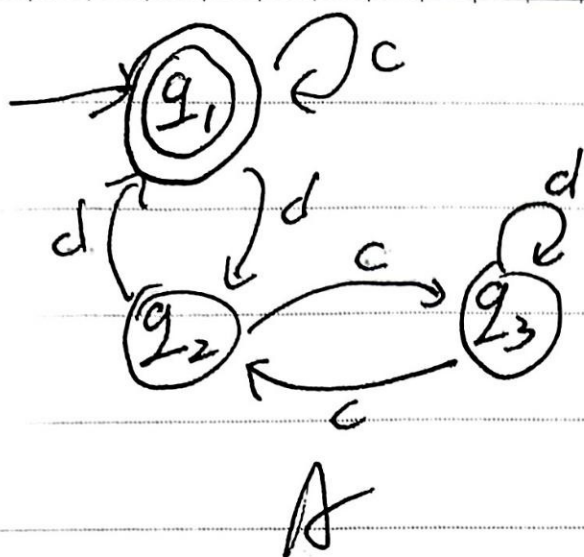
(2) If Initial State is Final State of one automaton, then in second automaton also Initial State must be Final State for them to be equivalent



Mo Tu We Th Fr Sa Su

Memo No. _____

Date ____/____/____



Startes	c	d
<u>(q1, q4)</u>	(q1, q4) FS FS	(q2, q5) IS IS
(q2, q5)	(q3, q6) IS IS	(q2, q4) FS FS
<u>(q3, q6)</u>	(q2, q7) IS IS	(q3, q6) IS IS ✓
(q2, q7)	<u>(q3, q6)</u> IS IS	<u>(q1, q4)</u> FS FS

no more new pair from table

⇒ A and B are equivalent

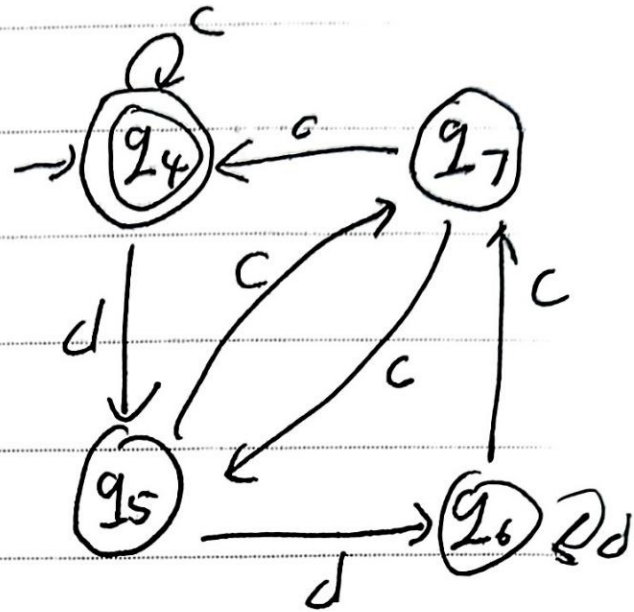
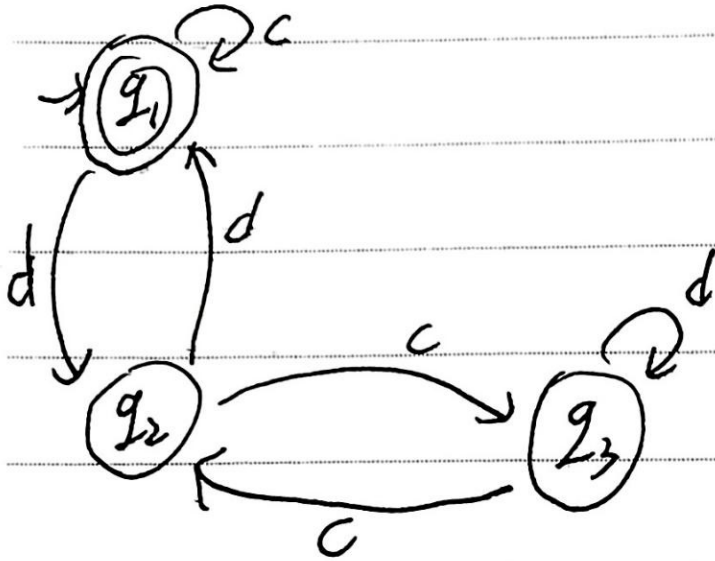


Mo	Tu	We	Th	Fr	Sa	Su
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Memo No. _____

Date / /

⑤ Example



A

states

c

d

(q_1, q_4)

(q_1, q_4)

(q_2, q_5)

$\downarrow$
FS FS ✓

$\downarrow$
IS IS ✓

(q_2, q_5)

(q_3, q_7)

✓

(q_1, q_6)

X

$\downarrow$
IS IS

$\downarrow$
FS IS

⇒ they are not equivalent automata

A ≠ B